



# Embedding models

Turning text, images, and other complex data into covariates

# A vector that captures **meaning**, not raw bytes

We want to feed rich data — a scan, a clinical note — into our models. The raw bytes are the wrong thing; we want a compact summary of **what the data means**:

## AN IMAGE

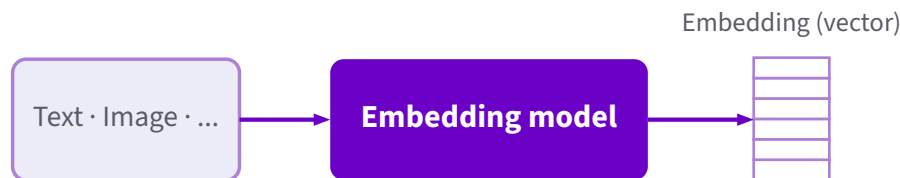
**not** the pixel intensities — but the objects present, their characteristics, what is happening, the style.

## A DOCUMENT

**not** the individual words — but the topic, the information conveyed, the sentiment, the language.

An **embedding model** is what turns the data into such a vector.

# A learned map from data to a meaningful vector



An **embedding model** is any (usually **pretrained**) map from data → a fixed-length vector whose **geometry is meaningful**.

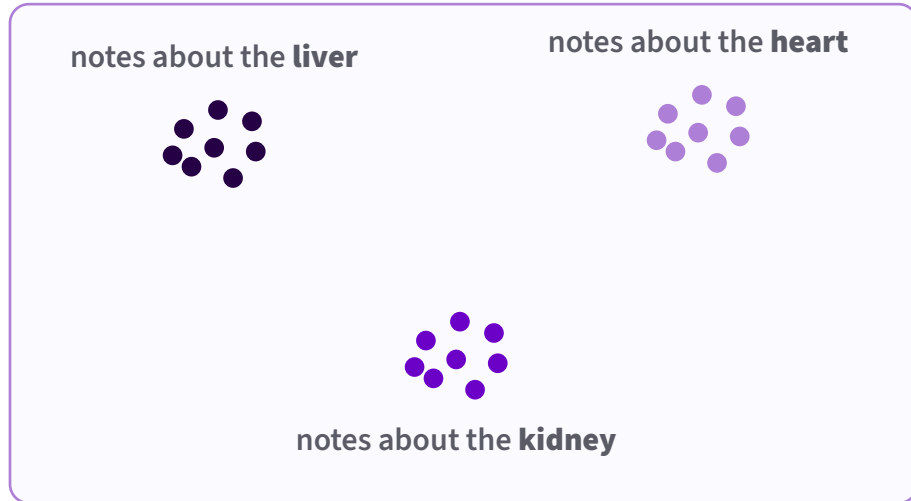
## A VAE ENCODER IS ONE EXAMPLE

The encoder from the generative story maps data → the **parameters** of its latent distribution; the posterior **mean** is an embedding.

## ...BUT EMBEDDINGS ARE BROADER

Most aren't generative at all: **no decoder**, trained **directly** for useful geometry — e.g. **contrastive** objectives on similar/dissimilar pairs — not for reconstruction.

# Similar inputs land close together

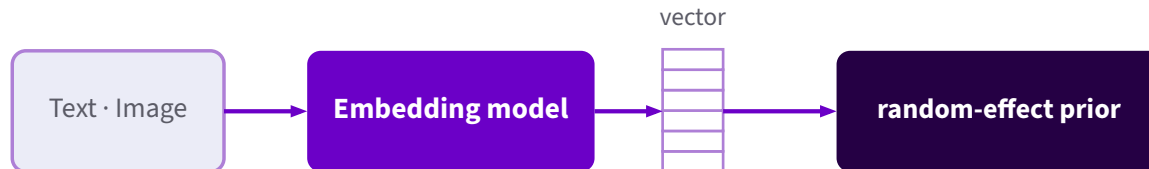


Distance in the vector space **means** something: semantically similar inputs sit **near each other**, unrelated ones far apart.

**And that geometry is the training signal:** the model is shown many pairs labelled **similar** vs **different**, and learns to pull the similar together and push the rest apart — no decoder, no reconstruction.

[huggingface.co/spaces/mteb/leaderboard](https://huggingface.co/spaces/mteb/leaderboard)

# A unified “API”: any modality → a vector covariate



## THE SMALL-N WIN

The model was **pretrained on far more data than our trial** — so we get strong features **without** training a big model on our own small dataset.

And heterogeneous data — images, notes, signals — all become **simple numeric vectors**: one interface for all of it.

## NEXT: USE IT AS A COVARIATE

Feed the embedding into the model — e.g. to **condition the random-effect prior** (augment). That is the **embeddings exercise**.